

MindMate

An AI-Powered Mental Health Assistant for Students

M.Sc. (Data Science), Final Project Report

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The Problem

Why MindMate?

- Student mental-health needs are rising, but counselling capacity is limited (WHO, 2022).
- Stigma, cost and waiting times stop many students seeking help early.
- Generic apps rarely personalise support or handle crises transparently.
- Need: accessible, stigma-free, personalised first-line support, distinct from clinical care.

Project Objectives

Three core objectives

- Design and develop an AI mental-health assistant for students.
- Personalise its support using machine-learning algorithms.
- Evaluate its effectiveness on mental-health and academic outcomes.

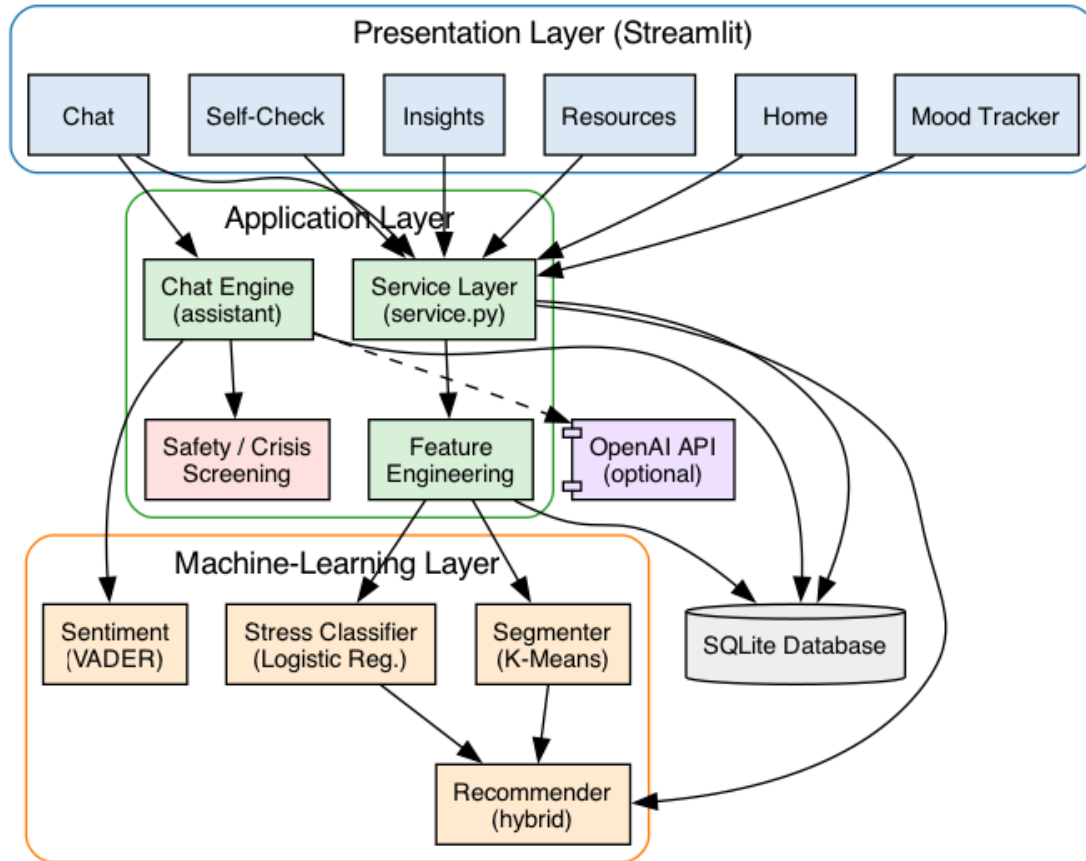
MindMate, Solution Overview

What we built

- Streamlit web app: Home, Chat, Mood Tracker, Self-Check, Insights, Resources.
- Empathetic assistant (OpenAI API with a robust offline fallback engine).
- ML personalisation: sentiment, student segmentation, stress prediction, recommendations.
- Safety-first: every message screened for crisis signals before any reply.
- PHQ-9 and GAD-7 self-checks with validated severity bands.

System Architecture

Figure: MindMate system architecture



- Presentation layer (Streamlit)
- Service layer orchestrates logic
- Assistant + ML domain layer
- Local SQLite persistence
- OpenAI API is optional

Technology Stack

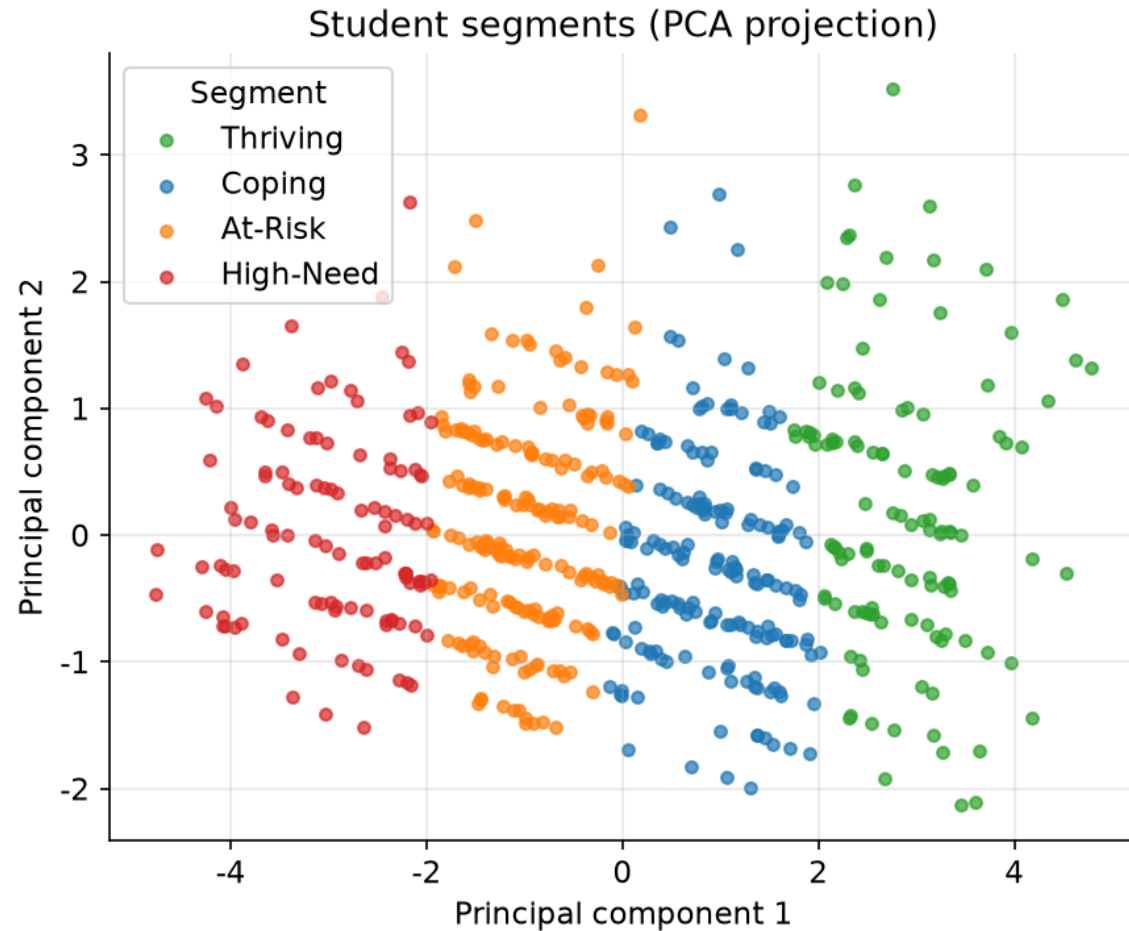
Open-source and reproducible

- Python 3.12, Streamlit (multipage UI).
- scikit-learn (K-Means, Logistic Regression, PCA); VADER sentiment.
- OpenAI Chat Completions API + offline rule-based engine.
- pandas, numpy, scipy; plotly and matplotlib for visualisation.
- SQLite persistence; joblib models; 34 pytest tests.

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Machine-Learning Personalisation

Student Segmentation (K-Means)



■ Segments: Thriving, Coping, At-Risk, High-Need

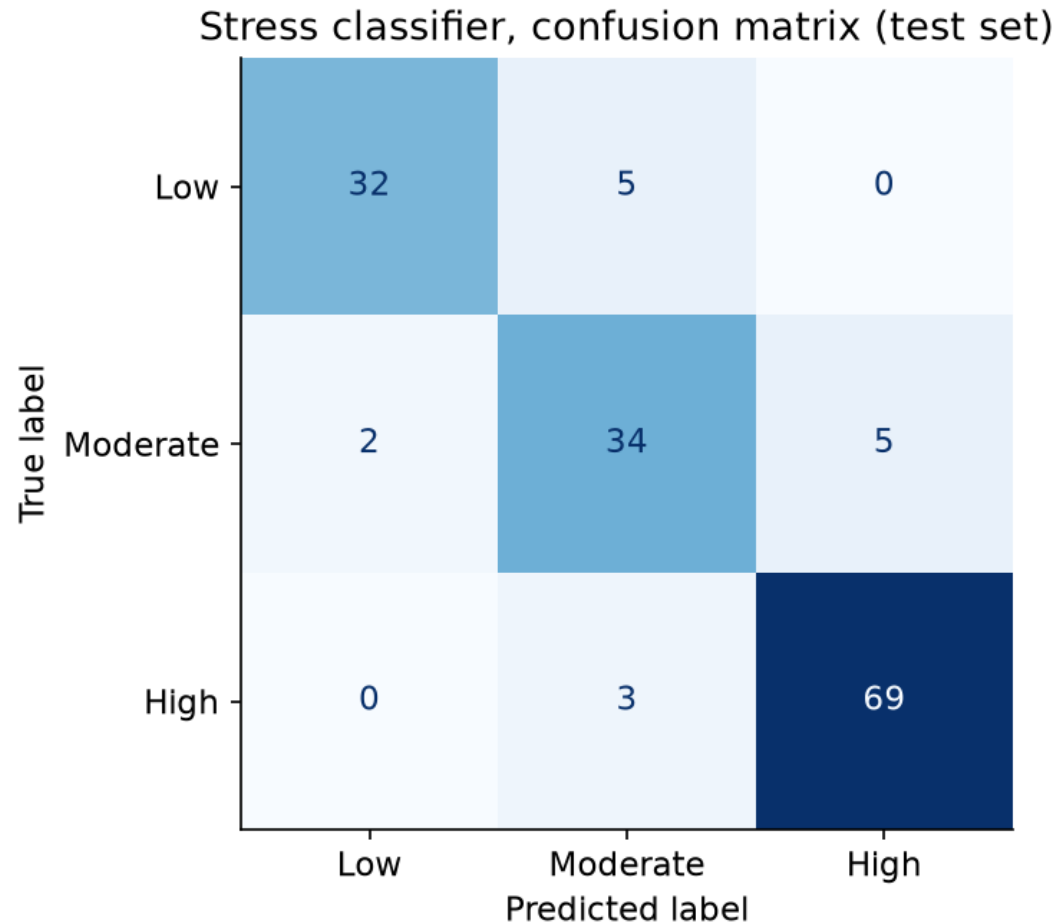
■ Six wellbeing features

■ Silhouette score 0.209

■ Labels ranked by wellbeing index

■ Each segment maps to a plan

Stress Classification



■ Logistic Regression (Low/Mod/High)

■ Test accuracy: 90.0%

■ Macro-F1: 0.889

■ Feeds dashboard + recommender

■ Transparent probabilities

Safety-First Design

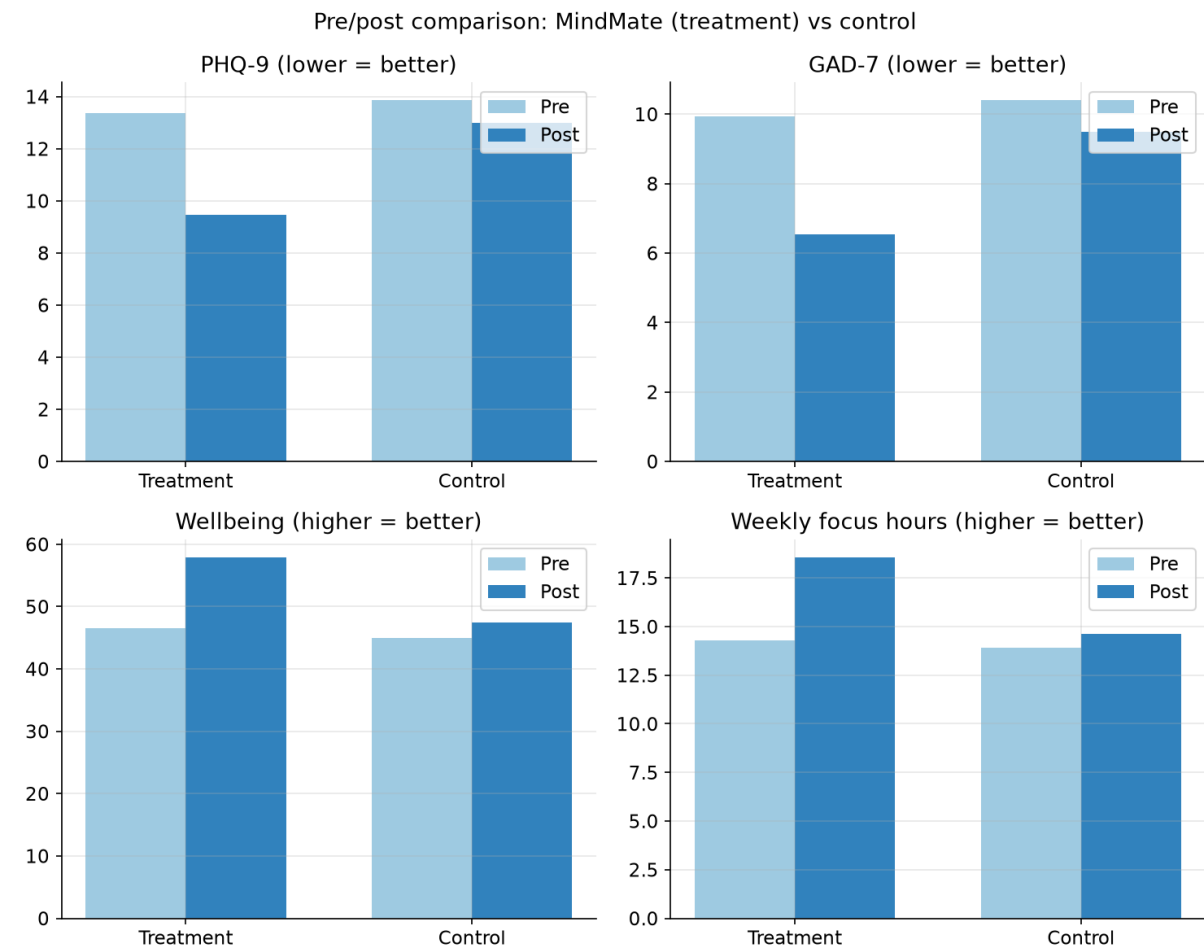
Ethics and user safety

- Every message screened for crisis signals BEFORE a reply is generated.
- Three risk levels: none, elevated, crisis.
- Crisis -> calm message + 24x7 helplines (Tele-MANAS, Vandrevalla, AASRA, iCall).
- Conservative by design: a missed crisis is far costlier than a false alarm.
- Never diagnoses; defers clinical concerns to professionals.

02

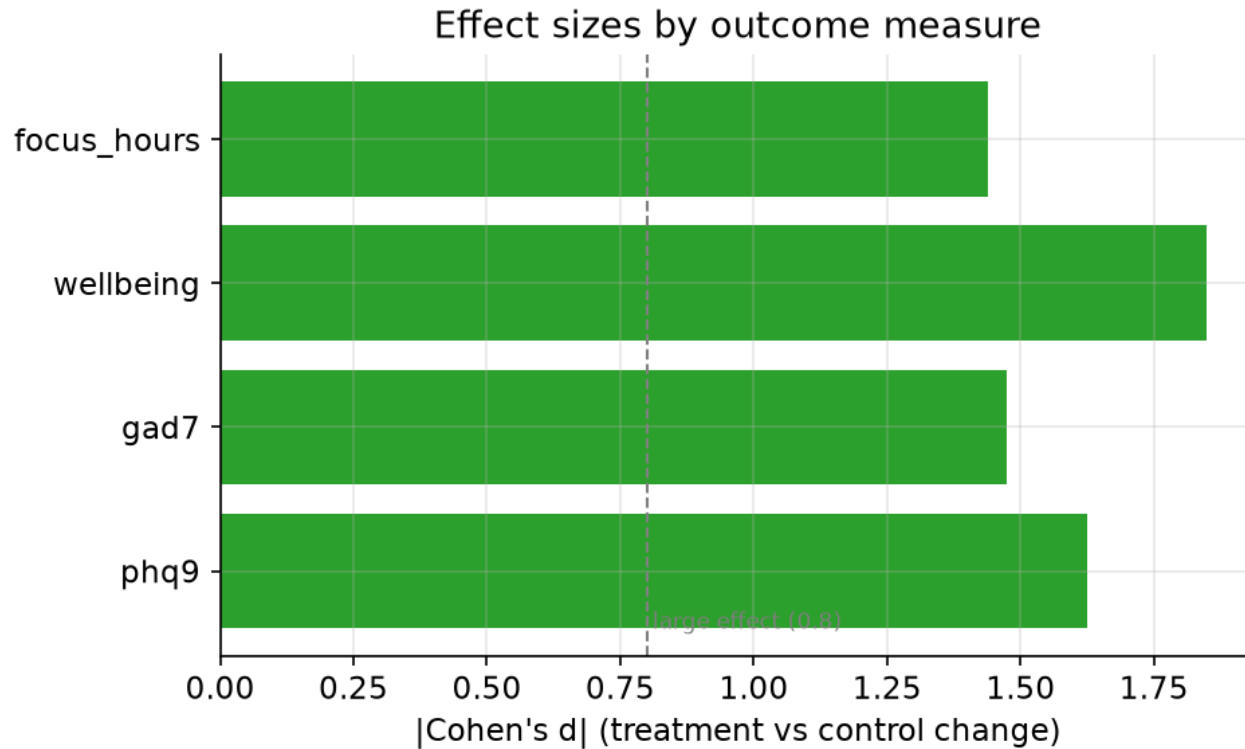
Evaluation of Effectiveness

Pilot Evaluation Results



Simulated 4-week pilot (120 treatment vs 120 control). Synthetic data, demonstrates the evaluation methodology, not clinical efficacy.

Effect Sizes



- PHQ-9: $d = 1.62$ (large)
- GAD-7: $d = 1.47$ (large)
- Wellbeing: $d = 1.85$ (large)
- Focus hours: $d = 1.44$ (large)
- All $p < 0.001$ (synthetic)

Testing & Validation

Quality assurance

- 34 automated pytest tests (unit, integration, system), all passing.
- Safety-critical tests (crisis detection; no over-flagging of 'killing it in exams').
- Hold-out validation: accuracy 0.900, macro-F1 0.889; silhouette 0.209.
- Reproducible end-to-end via a fixed random seed.

Conclusion & Future Scope

Outcomes and next steps

- All three objectives met: working assistant, ML personalisation, reproducible evaluation.
- Contributions: transparent personalisation, safety-first + offline design, reproducible pipeline.
- Future: real IRB-approved user study; multilingual support; mobile app.
- Future: transformer emotion models; counselling-referral integration; secure cloud deployment.

Thank You

Questions & Discussion

Project repository: github.com/gannyprodigy/mindmate-mental-health-assistant